

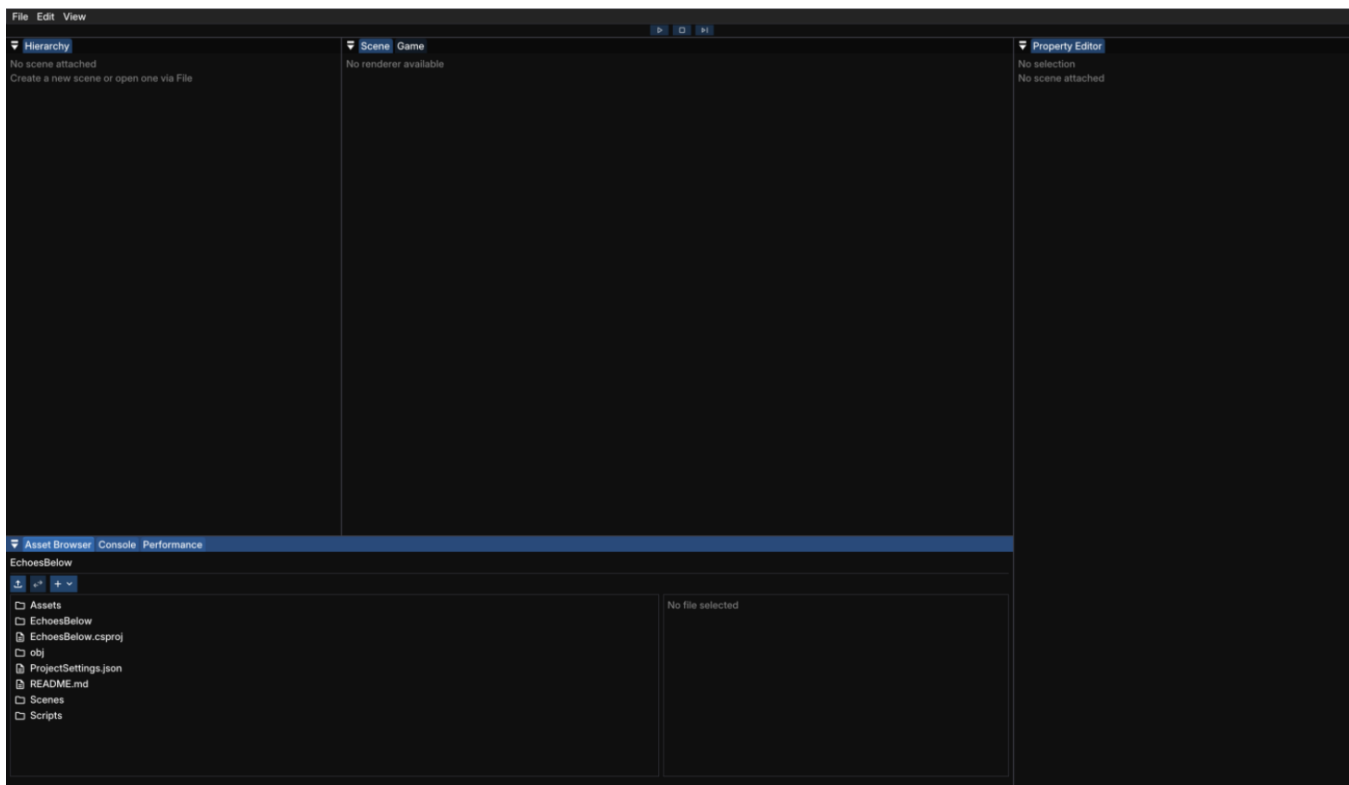
# GrapeEngine Level Editor Guide

Explains the usage of the editor panels (Hierarchy, Inspector, Asset Browser, Scene/Viewport, Playback Controls, Prefab Editor, Console) and common workflows like adding entities and assigning assets.

## Editor Layout & Docking

The editor uses ImGui docking to organize panels:

- Top toolbar: Game Controls (Play/Pause/Stop, Step)
- Left: Hierarchy
- Center: Scene (viewport) and Game (scene camera output)
- Right: Property Editor and Prefab Editor
- Bottom: Asset Browser, Console, Performance



## Playback Controls (Play/Pause/Stop)

- ♦ **Play:** Saves a snapshot and starts simulation
- ♦ **Pause:** Freezes time (timeScale = 0)
- ♦ **Stop:** Restores snapshot; returns to edit state
- ♦ **Step:** Advances one frame while paused

Hotkeys: **Ctrl+P** (Play/Stop), **Ctrl+Shift+P** (Pause/Resume), **Alt+P** (Step)

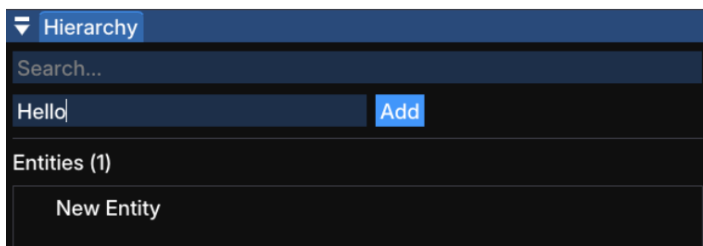


## Hierarchy Panel

Lists all entities with parent/child relationships. Supports selection, search/filter, context menu actions, reparenting (drag-drop), prefab instantiation, and creation.

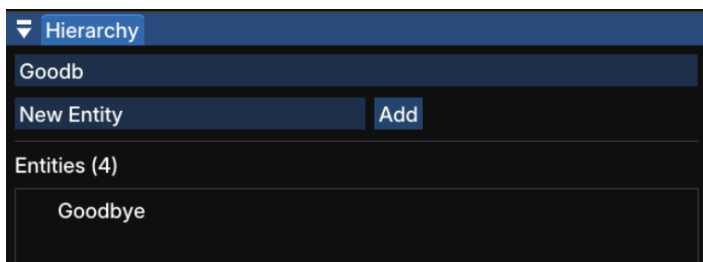
### Create a New Entity

1. Open **Hierarchy** panel.
2. Enter a name in the input field (defaults to "New Entity").
3. Click **Add** to create a root-level entity.
4. Drag the entity to reparent under another node.



### Search & Count

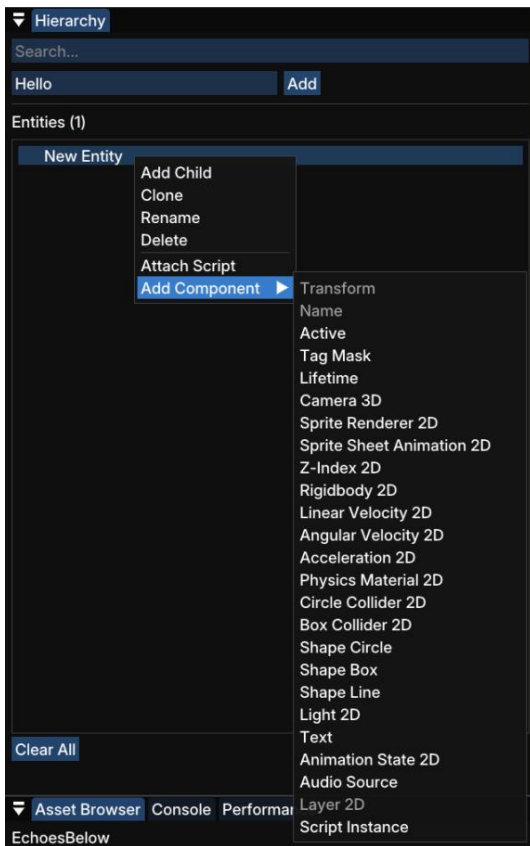
- Use the search field to filter by name.
- Entity count excludes hidden/system entities.



### Context Menu Actions

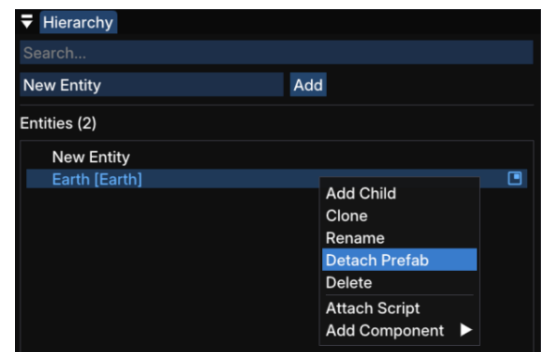
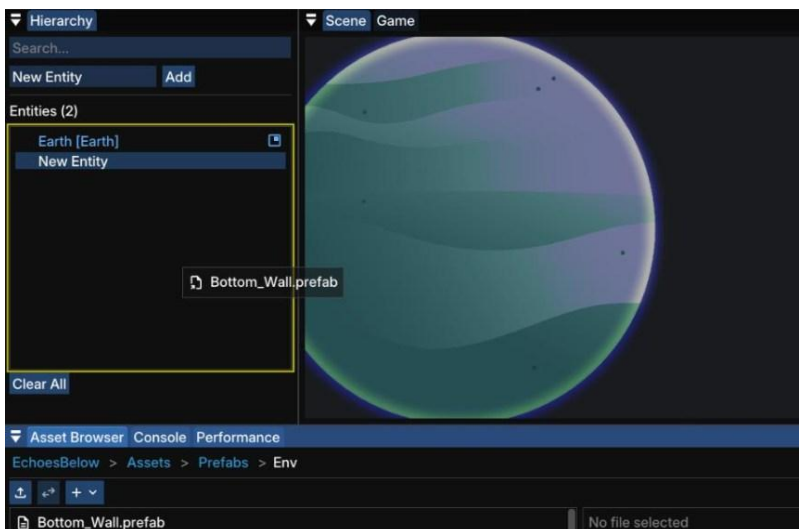
- **Add Child** (creates a new child entity under the selected parent)
- **Clone** (duplicates each selected entity, including its components and hierarchy)
- **Rename** (changes the name of the selected entity)
- **Detach Prefab** (breaks the link to the source prefab)

- **Delete** (removes the selected entity from the hierarchy list)
- **Attach Script / Detach Script** (Windows file dialog to import or remove scripts)
- **Add Component** submenu with available component types



## Prefab Instantiation

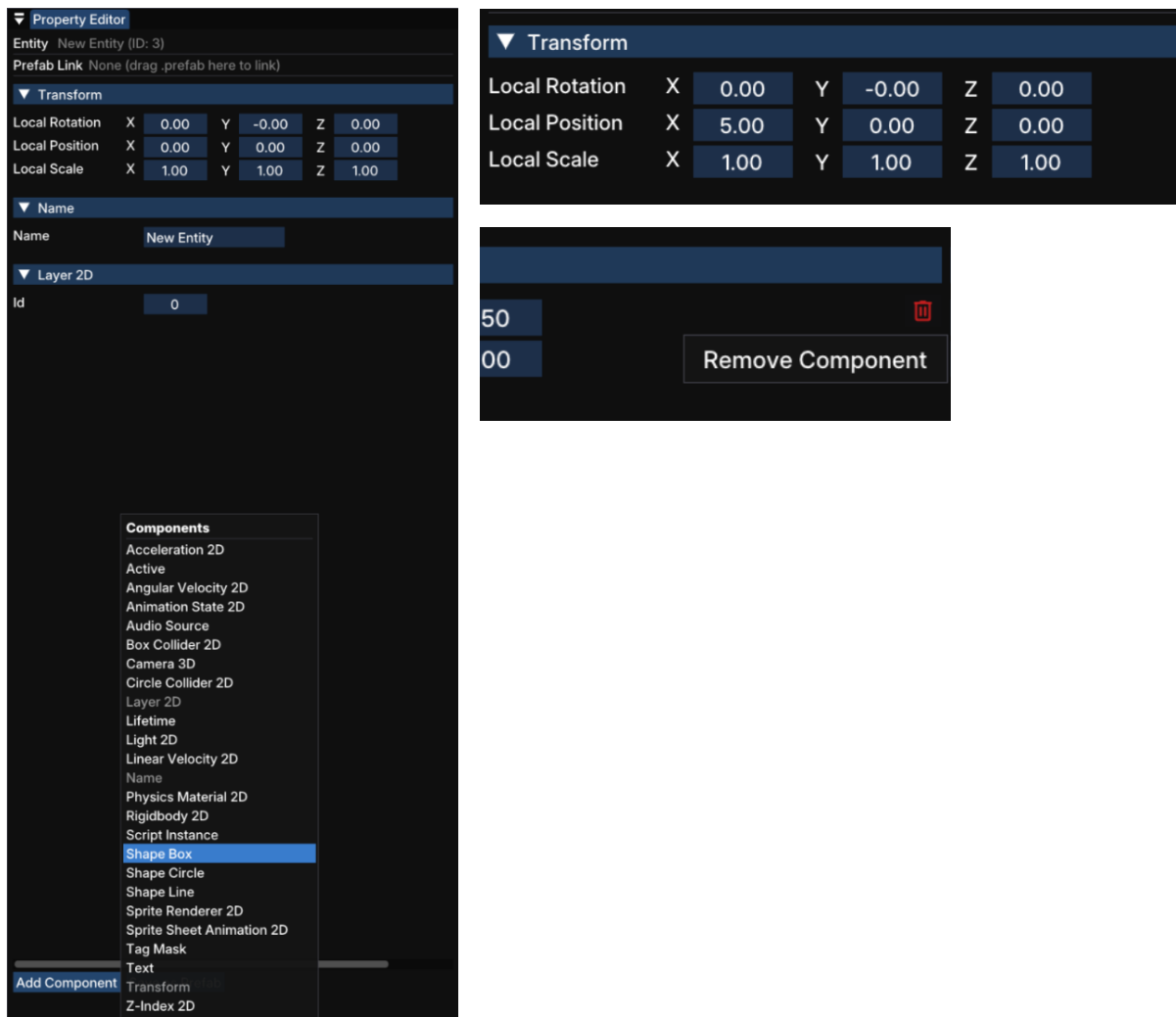
- Drag a `.prefab` onto a node to instantiate as a child.
- Instances are linked with a `PrefabLink` component and can be detached.



## Inspector (Property Editor)

Select an entity to edit its components. Use **Add Component** to add new ones. Edits apply immediately and are reflected in the scene; changes are applied back to components after the UI pass.

- ◆ Edit **LocalTransform** (Position, Rotation, Scale)
- ◆ Edit rendering, physics, and custom script components
- ◆ Undo records transform changes automatically
- ◆ **Add Component** popup sorted alphabetically
- ◆ Safe deletion via deferred remove (applied after UI loop)



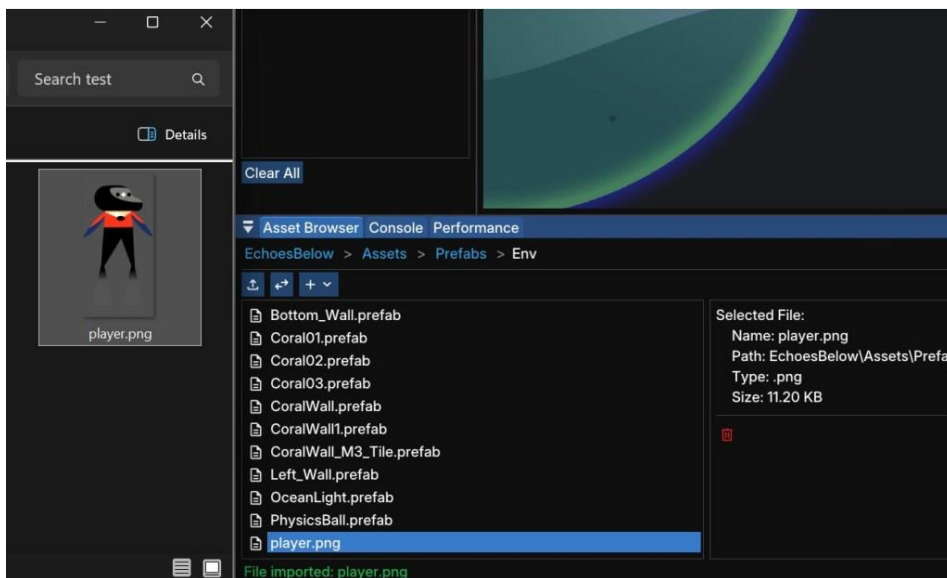
## Asset Browser

Browse assets, import new files, and drag-drop into components.

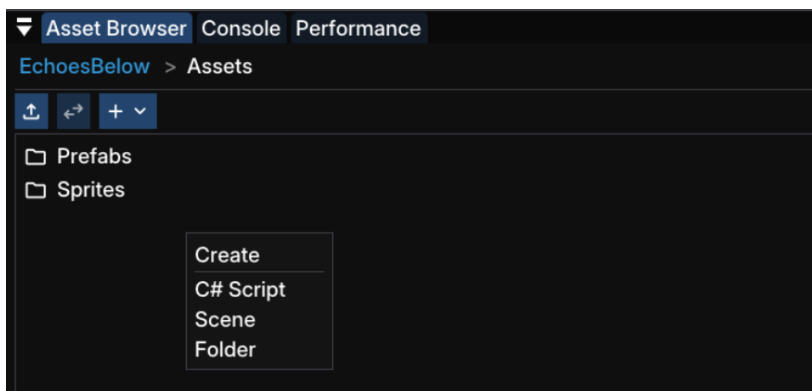
- **Internal drag-drop:** Drag a texture onto a SpriteRenderer field or audio onto AudioSource



- **External drop:** Drag files from Windows Explorer into the editor window to import



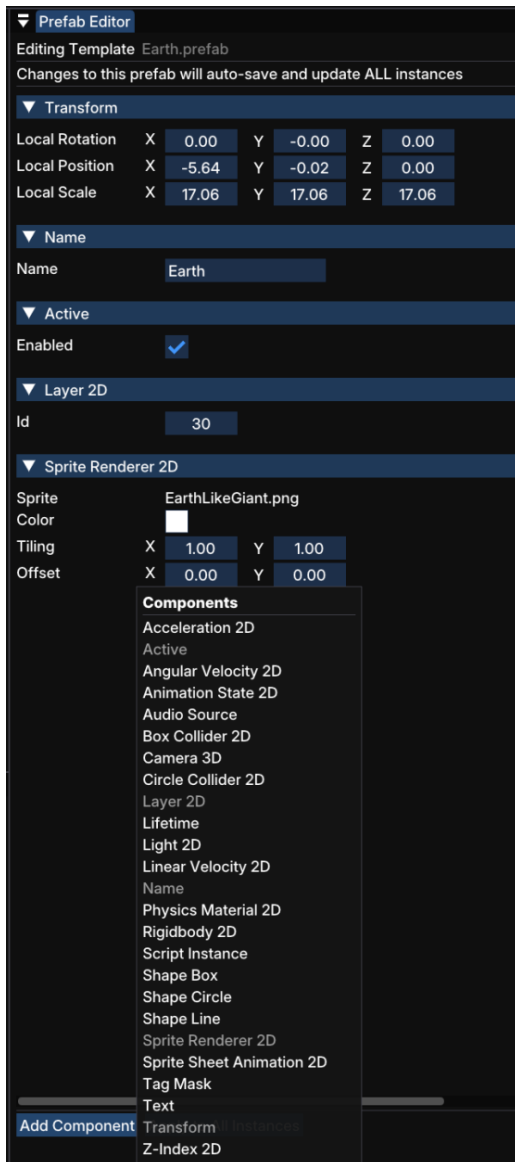
- Create assets: Right-click or use the **Create** menu (Script, Scene, Folder)
- Breadcrumb navigation at the top; multi-select with status feedback Context menu:
  - Rename (F2), Copy, Cut, Paste, Delete
  - Folder drop target: copy/move dragged assets; hold **Ctrl**
  - Prefabs: **Load Prefab** into scene, **Edit Prefab** in editor to copy



## Prefab Editor

Edit prefab components and save; apply changes to new and existing instances.

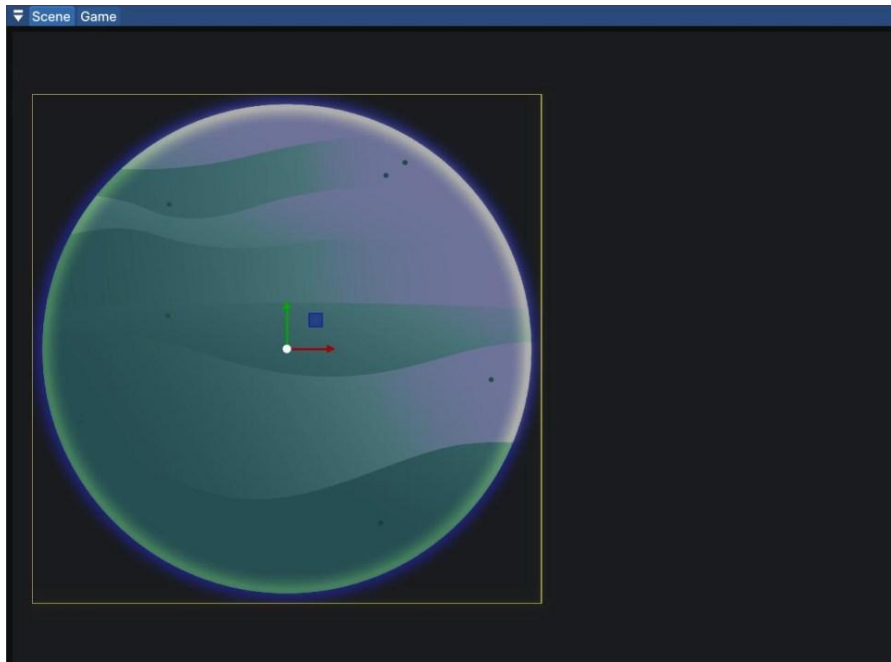
- ♦ Open a `.prefab` from Asset Browser and choose **Edit Prefab**
- ♦ Use **Add Component** and edit properties; changes auto-save
- ♦ Apply changes to all instances via prefab actions
- ♦ Components are displayed with Transform first, Name second, then alphabetical



## Scene / Viewport

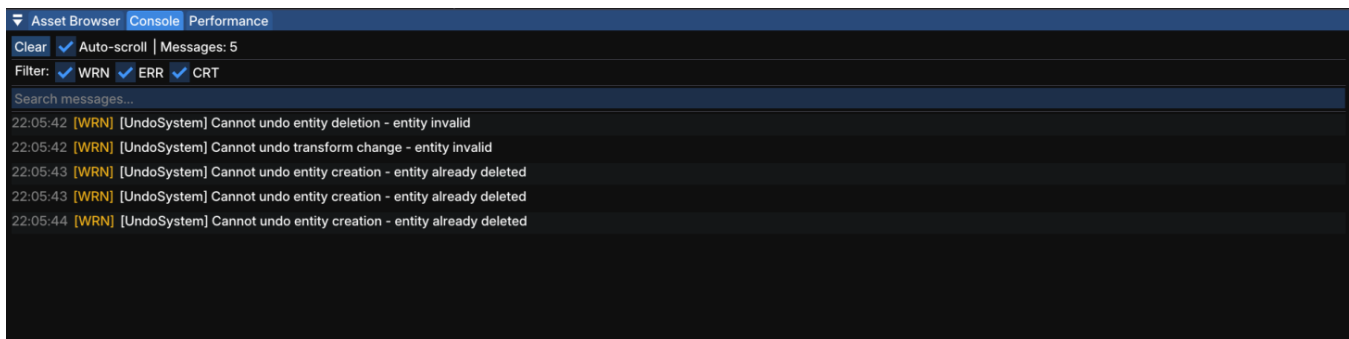
The **Scene** viewport shows the editor camera view; the **Game** tab shows the scene camera output.

- ◆ Resize the viewport; camera aspect updates automatically
- ◆ Selection in viewport syncs with Hierarchy
- ◆ Overlay shows FPS and flush count
- ◆ Tone-mapped LDR preview ensures colors match game output



## Console & Performance

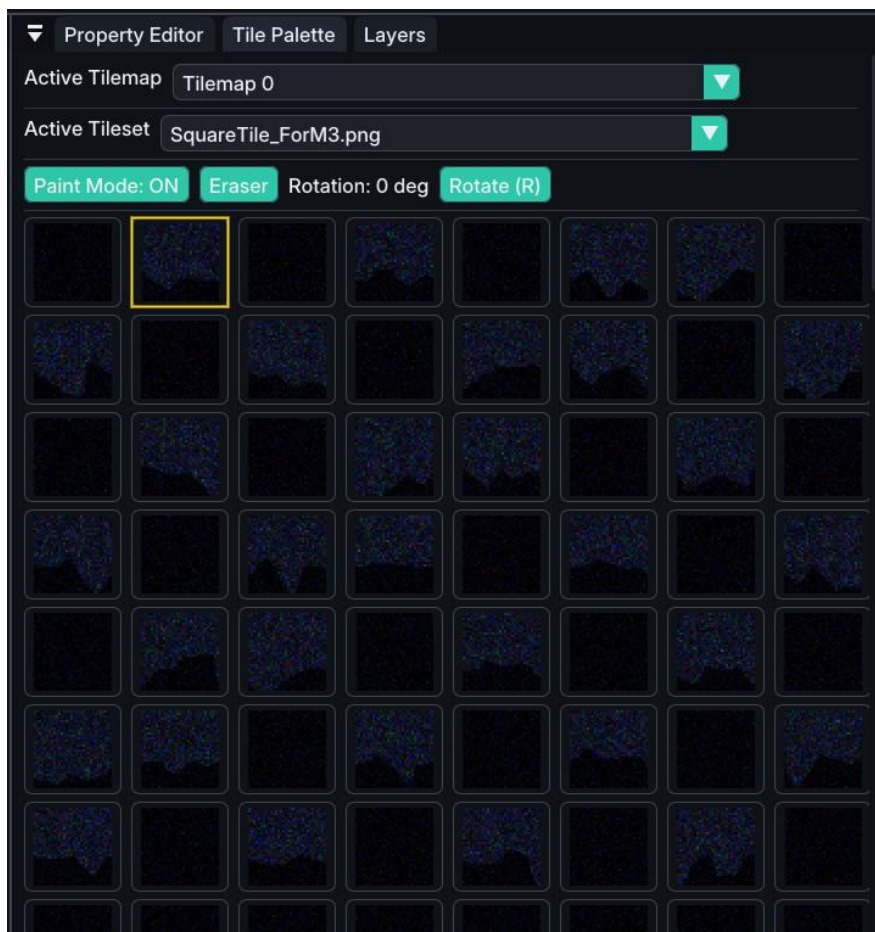
Use **Console** for logs and **Performance** for frame statistics.



## Tile Palette

The **Tile Palette** is an editor tool for painting tiles onto a tilemap in the game engine. What it does is it lets the user select tiles from a palette and paint them into the level by clicking in the viewport.

- ♦ **Select a tile:** Click a tile in the palette window to pick it
- ♦ **Paint:** Left-click in the viewport to place that tile
- ♦ **Erase:** Right-click in the viewport to delete tiles
- ♦ **Rotate:** Change the rotation (0-3) before painting if the tile needs to be rotated



## Common Workflow: Assign a Texture

1. Select an entity with `SpriteRenderer2D`.
2. In `Asset Browser`, select a `.png` file.
3. Drag the file onto the `Sprite` field in `Property Editor`.
4. Texture path and metadata are set; scene updates immediately.



## Tips

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- ♦ Use Play/Pause/Stop for real-time testing; Stop restores your edit state.
- ♦ Scenes/prefabs are JSON; assets can be added without recompiling.